T. Anderson Keller

Curriculum Vitæ

Email: t.anderson.keller@gmail.com
Website: akandykeller.github.io
Google Scholar: scholar.google.com/citations?user=Tb86kC0AAAAJ

Kempner Institute
Harvard University
Boston, MA 02134

Academic Experience

2023-cur **Kempner Research Fellow**, Kempner Institute, Harvard University, USA.

2018-2023 **Ph.D. Machine Learning**, Advisor: Max Welling; University of Amsterdam, The Netherlands.

2015-2017 **M.S. Computer Science**, Advisor: Garrison Cottrell; University of California San Diego, USA.

2011-2015 **B.S. Computer Science (Honors)**, Advisor: Yaser Abu-Mostafa; California Institue of Technology, USA.

Industry Experience

2022 **Research Intern**, Apple Machine Learning Research. Barcelona, Spain.

2016-2018 **Deep Learning Data Scientist**. Nervana → (acquired) → Intel AI Lab. San Diego, USA.

Teaching Experience

2025 **Course Instructor**: "The Neuroscience of Artificial Intelligence". Harvard College, Bachelor's.

2020 **Teaching Assistant**: "Machine Learning 1". University of Amsterdam, Bachelor's.

2019 **Teaching Assistant**: "Machine Learning 2". University of Amsterdam, Master's.

2016 **Head Teaching Assistant**: "Data Visualization". University of California San Diego, Master's.

Selected Advising Experience

2023-cur **Emma Finn**. Undergraduate, Harvard College. (Daily supervisor)
Outcome: Rhodes Scholarship Recipient '25 & Two Kempner Undergraduate Research Fellowships.

2024-cur **Mozes Jacobs**. Ph.D. Candidate, Harvard. (Daily supervisor during primary advisor's sabatical; last author)
Outcome: CCN '25 Proceedings (Oral), & ICLR '25 (top 7% scores).

2025-cur **Hansen Lillemark**. Ph.D. Candidate, UCSD. (Daily supervisor for summer research visit; last author)
Outcome: ICLR '25 (under review), Accepted submission at 2 NeurIPS '25, Workshops .

2020-2021 **Qinghe Gao**. Master's Candiate, University of Amsterdam. (Daily supervisor for master's thesis)
Outcome: Best Paper Award @ NeurIPS '25 SVRHM workshop.

Selected Publications

ICLR '26
Under Review
H. Lillemark, B. Huang, F. Zhan, Y. Du, and T. A. **Keller** (2026). "*Flow Equivariant World Modeling for Partially Observed Dynamic Environments*". In: *International Conference on Learning Representations (ICLR)*. Under review. url: https://openreview.net/forum?id=W7WUJTGByR

ICLR '26
Under Review
M. Jacobs, T. Fel, R. Hakim, A. Brondetta, D. E. Ba, and T. A. **Keller** (2026). "*Block Recurrent Dynamics in Vision Transformers*". In: *International Conference on Learning Representations (ICLR)*. Under review. url: https://openreview.net/forum?id=gH3HhnfWLC

ICLR '26
Under Review
J. Bertram, L. Dyballa, T. A. **Keller**, S. Kinger, and S. W. Zucker (2026). "*How Neural is a Neural Foundation Model?*" In: *International Conference on Learning Representations*. Under review. Accepted at Data on Brain & Mind Workshop @ NeurIPS 2025. url: https://openreview.net/forum?id=jUy7vFgoZf

NeurIPS '25
Spotlight
T. A. **Keller** (2025). "*Flow Equivariant Recurrent Neural Networks*". In: *Advances in Neural Information Processing Systems (NeurIPS)*. **Spotlight, Top 13% accepted**. arXiv: 2507.14793. url: https://openreview.net/forum?id=N1KPOlcN6P

NeurIPS '25
Y. Song, T. A. **Keller**, S. Brodjian, T. Miyato, Y. Yue, P. Perona, and M. Welling (2025). "*Kuramoto Orientation Diffusion Models*". In: *Advances in Neural Information Processing Systems (NeurIPS)*. url: https://arxiv.org/abs/2509.15328

NeurIPS '25	A. Karuvally, F. Nowak, T. A. **Keller**, C. A. Alonso, T. Sejnowski, and H. T. Siegelmann (2025). "*Bridging Expressivity and Scalability with Adaptive Unitary SSMs*". In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. url: https://openreview.net/forum?id=s4zitEu2R8
NeurIPS '25 Workshop	S. Bharthulwar, T. A. **Keller**, M. Theodosis, and D. E. Ba (2025). "*From Extrapolation to Generalization: How Conditioning Transforms Symmetry Learning in Diffusion Models*". In: *NeurIPS 2025 Workshop on Symmetry and Geometry in Neural Representations*. url: https://openreview.net/forum?id=UI82R3lwar
CCN '25 Oral, Proceedings	M. Jacobs, R. C. Budzinski, L. Muller, D. E. Ba, and T. A. **Keller** (2025). "*Traveling Waves Integrate Spatial Information Through Time*". In: *Conference on Cognitive Computational Neuroscience (CCN)*. **Oral presentation, Top 7%**. url: https://openreview.net/forum?id=QEzqo546V5
Springer '25 Book	Y. Song, T. A. **Keller**, N. Sebe, and M. Welling (May 2025). *Structured Representation Learning*. en. Synthesis Lectures on Computer Vision. Cham, Switzerland: Springer International Publishing. url: https://link.springer.com/book/10.1007/978-3-031-88111-4
COSYNE '25 Abstract	T. A. **Keller** (2025). "*Nu-Wave State Space Models: Traveling Waves as a Biologically Plausible Context*". In: *Science Communications Worldwide*. doi: 10.57736/b30b-8eed. url: https://www.world-wide.org/cosyne-25/nu-wave-state-space-models-traveling-3803805f
AISTATS '26 Under Review	E. L. Byrnes Finn, B. Wang, T. A. **Keller**, and D. E. Ba (2026). "*Where the Score Lives: A Wavelet View of Diffusion*". In: *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*. Under review. Accepted at SPIGM Workshop @ NeurIPS 2025.
PNAS '25	L. H. B. Liboni, R. C. Budzinski, A. N. Busch, S. Löwe, T. A. **Keller**, M. Welling, and L. E. Muller (2025). "*Image segmentation with traveling waves in an exactly solvable recurrent neural network*". In: *Proceedings of the National Academy of Sciences (PNAS)* 122.1, e2321319121. doi: 10.1073/pnas.2321319121. url: https://www.pnas.org/doi/abs/10.1073/pnas.2321319121
ICML '25 Workshop	E. L. B. Finn, T. A. **Keller**, M. Theodosis, and D. E. Ba (2025). "*Origins of Creativity in Attention Based Diffusion Models*". In: *High-dimensional Learning Dynamics 2025 @ ICML '25*. url: https://openreview.net/forum?id=xMqRYKvrD1
CCN '25 Abstract	Y. Song, T. A. **Keller**, Y. Yue, P. Perona, and M. Welling (2024). "*Langevin Flows for Modeling Neural Latent Dynamics*". In: *Conference on Cognitive Computational Neuroscience (CCN)*. url: https://arxiv.org/abs/2507.11531
ICLR '24	T. A. **Keller**, L. Muller, T. Sejnowski, and M. Welling (2024). "*Traveling Waves Encode the Recent Past and Enhance Sequence Learning*". In: *International Conference on Learning Representations (ICLR)*. url: https://openreview.net/forum?id=p4S5Z6Sah4
arXiv '24	T. A. **Keller**, L. Muller, T. J. Sejnowski, and M. Welling (2024). *A Spacetime Perspective on Dynamical Computation in Neural Information Processing Systems*. arXiv: 2409.13669 [q-bio.NC]. url: https://arxiv.org/abs/2409.13669
TPAMI '24	Y. Song, T. A. **Keller**, Y. Yue, P. Perona, and M. Welling (2024). "*Unsupervised Representation Learning from Sparse Transformation Analysis*". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence*. arXiv: 2410.05564 [cs.LG]. url: https://arxiv.org/abs/2410.05564
arXiv '24	E. Finn, T. A. **Keller**, E. Theodosis, and D. E. Ba (2024). *Learning Artistic Signatures: Symmetry Discovery and Style Transfer*. arXiv: 2412.04441 [cs.CV]. url: https://arxiv.org/abs/2412.04441
CCN '24 Abstract	T. A. **Keller**, T. Konkle, and C. Conwell (2024). "*Towards the Use of Relative Representations for Lower-Dimensional, Interpretable Model-to-Brain Mappings*". In: *Conference on Cognitive Computational Neuroscience (CCN)*. url: https://2024.ccneuro.org/poster/?id=317
ICML '23	T. A. **Keller** and M. Welling (2023). "*Neural Wave Machines: Learning Spatiotemporally Structured Representations with Locally Coupled Oscillatory Recurrent Neural Networks*". In: *Proceedings of the 40th International Conference on Machine Learning (ICML)*. vol. 202. Proceedings of Machine Learning Research, pp. 16168–16189. url: https://proceedings.mlr.press/v202/keller23a.html
NeurIPS '23	Y. Song, T. A. **Keller**, N. Sebe, and M. Welling (2023). "*Flow Factorized Representation Learning*". In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 36. Curran Associates, Inc., pp. 49761–49782. url: https://proceedings.neurips.cc/paper_files/paper/2023/file/9bfc2c20fa2f56a18397eafe1be8a50a-Paper-Conference.pdf

TMLR '23 — T. A. **Keller**, X. Suau, and L. Zappella (2023). "*Homomorphic Self-Supervised Learning*". In: *Transactions on Machine Learning Research*. issn: 2835-8856. url: https://openreview.net/forum?id=tEKqQgbwbf

ICML '23 — Y. Song, T. A. **Keller**, N. Sebe, and M. Welling (2023). "*Latent traversals in generative models as potential flows*". In: *Proceedings of the 40th International Conference on Machine Learning*. ICML'23. Honolulu, Hawaii, USA: JMLR.org. url: https://dl.acm.org/doi/10.5555/3618408.3619746

ICML '23 — X. Suau, F. Danieli, T. A. **Keller**, A. Blaas, C. Huang, J. Ramapuram, D. Busbridge, and L. Zappella (2023). "*DUET: 2D Structured and Approximately Equivariant Representations*". In: *Proceedings of the 40th International Conference on Machine Learning*. Vol. 202. Proceedings of Machine Learning Research. PMLR, pp. 32749–32769. url: https://proceedings.mlr.press/v202/suau23a.html

NeurIPS '23 Workshop — N. L. Masclef and T. A. **Keller** (2023). *Deep Generative Models of Music Expectation*. arXiv: 2310.03500 [cs.SD]. url: https://arxiv.org/abs/2310.03500

COSYNE '22 Abstract — T. A. **Keller** and M. Welling (2022). "*Locally Coupled Oscillator Networks Learn Traveling Waves and Topographic Organization*". In: url: https://akandykeller.github.io/papers/LocoRNN.pdf

ICML '21 — T. A. **Keller**, J. W. T. Peters, P. Jaini, E. Hoogeboom, P. Forré, and M. Welling (July 2021). "*Self Normalizing Flows*". In: *Proceedings of the 38th International Conference on Machine Learning (ICML)*. vol. 139. Proceedings of Machine Learning Research. PMLR, pp. 5378–5387. url: https://proceedings.mlr.press/v139/keller21a.html

ICCVW '21 Oral — T. A. **Keller** and M. Welling (2021). "*Predictive Coding with Topographic Variational Autoencoders*". In: *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*. **Oral presentation**, pp. 1086–1091. doi: 10.1109/ICCVW54120.2021.00127

SVRHM '21 Best Paper Award — T. A. **Keller***, Q. Gao, and M. Welling (2021). "*Modeling Category-Selective Cortical Regions with Topographic Variational Autoencoders*". In: *Shared Visual Representations in Humans and Machines (SVRHM) Workshop @ NeurIPS*. **Best Paper Award**. url: https://openreview.net/forum?id=yGRq_lW54bI

NeurIPS '21 — T. A. **Keller** and M. Welling (2021). "*Topographic VAEs Learn Equivariant Capsules*". In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 34. Curran Associates, Inc., pp. 28585–28597. url: https://proceedings.neurips.cc/paper/2021/file/f03704cb51f02f80b09bffba15751691-Paper.pdf

Invited Talks

2025 **NSF IAIFAI Colloquium** (MIT/Harvard, USA): "Flow Equivariance: Enforcing Time-Parameterized Symmetries in Sequence Models". Recording: youtube.com/watch?v=fO-wsJgGeL8

2025 **Fields Symposium on Spatiotemporal Neural Dynamics** (U of T, CA): "The Computational Inductive Biases of Spatiotemporal Artificial Neural Networks". Recording: youtube.com/watch?v=Rd-4lubEYc4

2025 **Neuro AI Emerging Leaders Webinar**: "The Computational Inductive Biases of Spatiotemporal Dynamics". Recording: youtube.com/watch?v=nmiGZo5uPH4

2024 **NeurReps International Speaker Series**: "A Spacetime Perspective on Neural Information Processing Systems". Recording: youtube.com/watch?v=FeVHfmRSicQ.

2024 **Active Inference Institute Seminar** (Virtual): "Natural Structure in Artificial Nerual Networks". Recording: youtube.com/watch?v=1uIRljnLtc4.

2025 **AI, Brain, & Cognition Seminar, Center for NeuroImaging Research, Korea** (Virtual): "Computational Inductive Biases of Spatiotemporal Dynamics".

2025 **Guest Lecture: Applied Math 220, Geometric Deep Learning @ Harvard** (Harvard, USA): "Flow Equivariance".

2024 **Yale Wu Tsai Neuro AI Seminar**: "A Spacetime Perspective on Neural Information Processing Systems".

2024 **Woods Hole Computational Neuroscience Workshop** (Telluride, USA): "Wave Information Processing Systems".

2024 **CERN CMS Machine Learning Seminar** (Geneva, Switzerland): "Traveing Waves in Brains and Machines".

2024 **Computational Neuroscience Next Generation Symposium** (WashU, USA): "A Spacetime Perspective on Neural Information Processing Systems".

2023 **Frankfurt Institute for Advanced Studies, Giersch School and International Conference on Complex Systems** (Frankfurt, Germany): "Generalized Equivariance through Fluid Representations in Brains and Machines".

2023 **A.D. de Groot Cognitive AI Lecture** (Amsterdam, Netherlands): ”Topographic Organization and Traveling Waves as Natural Representational Structure for Artificial Neural Networks”.

2023 **Workshop on Structured Learning, Chalmers AI Research Center** (Gothenburg, Sweden): ”Natural Neural Structure for Artificial Intelligence”.

2022 **Seminar on Advances in Probabilistic Machine Learning** (Helsinki, Finland): ”Topographic Variational Autoencoders Learn Equivariant Capsules”.

Awards

2025 **Spotlight (Top 13% Accepted)**.
”Flow Equivariant Recurrent Neural Networks”, Keller, T. A., NeurIPS ’25.

2025 **Contributed Talk (Top 7% Accepted)**.
”Traveling Waves Integrate Information over Space”, Jacobs, M., ..., Keller, T. A., CCN ’25.

2025 **Top Reviewer Award**.
NeurIPS ’25. Award Value: $800.

2025 **Contributed Talk (Top 17%)**.
”Flow Equivariant Cybernetics”, Lillemark, ..., Keller, T. A., Champalimaud Foundation Research Symposium ’25.

2025 **Travel Award**.
Chamalimaud Foundation. Amount: $800.

2025 **FENS Brain Conference Travel Grant**.
FENS. Amount: $800.

2024 **Contributed Talk (Top 10%)**.
”Traveling Waves in State Space Models”. Keller, T. A.. From Neuroscience to AI Systems (NAISys) Conference.

2023 **Travel Award**.
COSYNE. Amount: $800.

2022 **Travel Award**.
CIFAR Neuroscience of Consciousness Winter School. Award Value: $1,500.

2021 **Contributed Talk (Top 10%)**.
”Predictive Coding with Topographic VAEs”. Keller, T. A. & Welling, M., Visual Inductive Priors Workshop, ICCV ’21.

2021 **Best Paper Award**.
”Modeling Catagory-Selective Cortical Regions with TVAEs”. Keller, T. A.*, Gao, Q.*, & Welling, M., Shared Visual Representations in Humans and Machines (SVRHM) Workshop; NeurIPS ’21. Award Value: $300.

Community Service

2025-cur **Workshop Co-organizer**, Teluride Neuromorphic Engineering Workshop on Spatiotemporal Dynamics

2025-cur **Speaker Series Organizer**, Foundation Role Models Speaker Series, CRISP Lab, Harvard

2025-cur **Reviewer**, ICLR Workshop Proposals

2025-cur **Reviewer**, TMLR Proceedings

2025-cur **Reviewer**, UAI Proceedings

2025-cur **Reviewer**, AAAI Proceedings

2025-cur **Reviewer**, CCN Proceedings

2024-cur **Reviewer**, NeurIPS Proceedings

2024-cur **Reviewer**, Neural Computation

2024-cur **Reviewer**, ICLR Proceedings

2024-cur **Reviewer**, AIStats Proceedings

2025-cur **Program Committee (Reviewer)**, Re-Align Workshop

2024-cur **Program Committee (Reviewer)**, NeuReps Workshop

2019-2022 **Board Member**, Inclusive AI, University of Amsterdam, ivi.fnwi.uva.nl/ellis/inclusive-ai/